Kaleb Rice

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Education

Master of Science in Mechanical Engineering – Portland State University — Starting September 2027

Bachelor of Science in Mechanical Engineering – Portland State University — September 2022 – Ongoing

GPA: 3.56 **Expected Graduation:** June 2027

Relevant Courses: Finite Element Analysis (FEA) · Materials Failure Analysis · Heat Transfer and Thermodynamics · Fluid Mechanics · Vibrations & System Dynamics · Manufacturing Processes

Technical Skills

CAD & Design: Creo, SolidWorks (CSWA), OnShape (COP), Fusion 360, Siemens Solid Edge, FreeCAD, KiCad

Fabrication: FDM & SLA 3D Printing, MasterCAM, Mill, Lathe, MIG / TIG Welding

Software & Coding: MATLAB, C++, Python, Arduino / ESP32 Microcontrollers

Additional: Design for Manufacturing (DFM), Geometric Dimensioning and Tolerancing (GD&T)

Experience

Machine Shop Technician, PSU Mechanical Engineering Lab Team (MELT) — October 2025 – Ongoing

- Trained students of various skill levels on safe operation of mills, lathes, welders, and other shop tools.
- Designed and fabricated equipment for raw material processing to improve production efficiency.

Automation Engineering Intern (MECOP), Epson Portland Inc. — April 2026 – September 2026

- Designed a modular adhesive applicator that reduced production downtime by 20 minutes each day.
- Managed the design and procurement of a cleanroom conveyor line to enable automatic packaging of flexible ink bags.
- Worked across departments to collect data and effectively implement various assembly line modifications to improve worker safety, efficiency, and quality of life.

Mechanical Design Engineering Intern (MECOP), Benchmade Knife Company — April 2025 – September 2025

- Designed 11 production-ready knife assemblies in Creo from concept to manufacturing, integrating DFM and material analysis principles.
- Researched and prototyped new compliant mechanisms, surface finishes, and manufacturing techniques to improve reliability and aesthetics.
- Created detailed engineering drawings to ASME Y14.5 standards to guide production and control quality.

Lead Mechanical Engineer, Portland State Aerospace Society — February 2023 – March 2025

- Directed mechanical design for a $200k open-source satellite communications station (UniClOGS), collaborating with multidisciplinary teams.
- Modeled structural components and performed CFD-based wind-load simulations to validate city permitting calculations.
- Trained peers in CAD, documentation, and design review practices to enhance collaboration and design manufacturability.

Freelance Engineering, Kale Creative Works — December 2021 – October 2023

- Provided nationwide CAD, 3D printing, CNC programming, and prototyping services for individuals and small businesses.
- Managed full client lifecycle: concept development, design consultation, manufacturing coordination, and assembly.

Projects & Leadership

Open-Source Plastic Scanner — October 2024 – March 2025

- Awarded $1,000 PSU Cleantech Challenge grant to prototype a near-infrared spectroscopy system for plastic recycling and filament production.

High School Makerspace Founder — May 2022 – April 2026

- Established and led a community makerspace offering STEAM education and fabrication tools; annual exhibitor at the Portland Winter Light Festival.

FIRST Robotics — September 2018 - February 2020

- Led the mechanical design of an omnidirectional drive robot and oversaw competition strategy.
- Returned as a volunteer judge in 2024 to supervise competitions and interview student teams.